Intraday Market Return Forecasting from the Factor Zoo

https://github.com/ChuaoD/intraday-market-return-forecasting

Can the recent returns of 272 stock portfolios predict the US stock market's return over the next 15 minutes, and is that prediction worth trading?

Overview

This project forecasts the **next 15-minute return of the US stock market** from the recent 15-minute returns of **272 portfolios**, using as forecasting models three regularized linear models: Ridge, Lasso and Elastic Net. Every forecast is made only from information available at the time it is made (avoiding look-ahead bias). The models are tested on **111,084 forecasts over 17 years (2004–2020)**. The project then asks four follow-up questions:

1. Are the forecasting accuracy differences among models statistically reliable?
2. Do the 272 portfolios help beyond the market's own past?
3. Could the forecasts make money after trading costs?
4. When, and in which market conditions, are the forecasts accurate?

The forecasting experiment partially reproduces Aleti, Bollerslev and Siggaard (2025, *Management Science*). The uncertainty analysis, predictor and regularization checks, trading evaluation and regime analysis are my own work (details).

Results at a glance

Brackets show 95% confidence intervals. R^2, basis points (bp) and the Sharpe ratio are explained in Section 1.

Question	Answer
1. Are the accuracy differences among models statistically reliable?	**All three models beat a zero forecast, but none clearly beats the others.** Ridge has the highest out-of-sample R^2, **0.171%** [0.102%, 0.239%]. Its lead over Lasso is 0.020 percentage points [−0.011, 0.050], and this interval includes zero.
2. Do the 272 portfolios help beyond the market's own past?	**Yes.** Ridge with all 816 predictors beats a Ridge model that uses only the market's own past returns by **0.096 percentage points** of R^2 [0.038, 0.159]. It still wins, by 0.108 [0.046, 0.176], when that simpler model is allowed weaker penalties.
3. Could the forecasts make money after trading costs?	**Only barely.** A simple long/short rule has a **Sharpe ratio of 2.15** [1.71, 2.58] before costs, but it trades so often that it breaks even at a cost of **0.74 bp** per unit traded. Trading only the strongest forecasts raises this to **1.10 bp**. Realistic costs for SPY are around 0.14–0.7 bp plus fees.
4. When, and in which market conditions, are the forecasts accurate?	**On high-volatility days.** These days produce all of the advantage of the 816-predictor model. The advantage is largest in 2008–2010 but is still present after 2010 (0.032 percentage points [0.004, 0.062]).

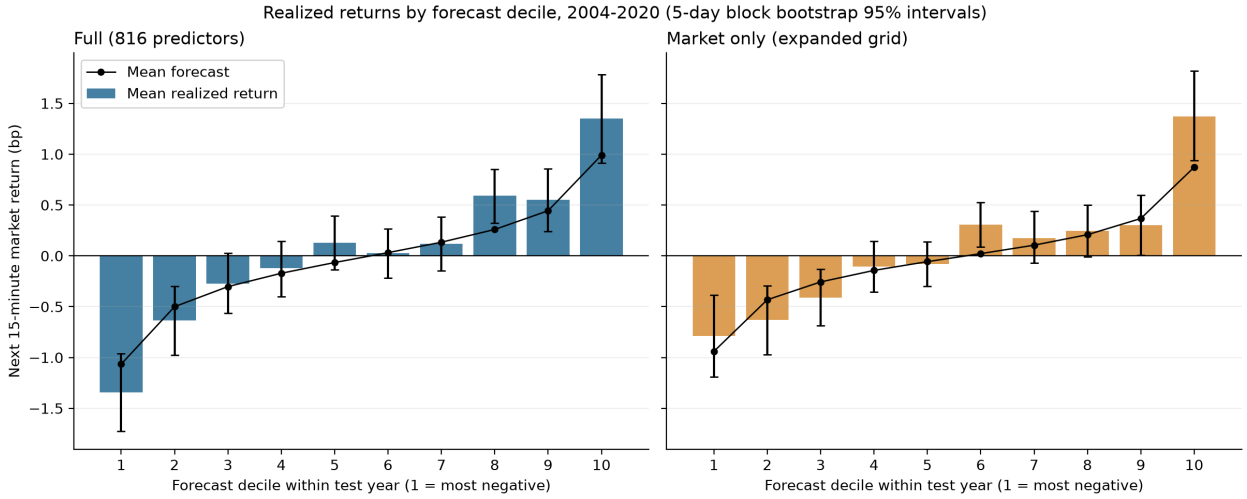


Figure 1: Mean realized return by forecast decile. Each test year's forecasts are sorted from most negative to most positive and split into ten equal groups (deciles). The bars show the average of the subsequent market returns for each group. Bars rising from left to right mean that higher forecasts tend to be followed by higher returns.

What I built

Reproduced from the paper

- **Data pipeline.** Converted 2.45 million one-minute timestamps for 272 portfolios (1996–2020) into 15-minute returns. Overnight returns are kept separate, and shortened trading days are kept as observed, without filling missing intervals.
- **Descriptive statistics.** Reproduced the paper's summary table of return, volatility, jump and correlation statistics for each portfolio group, its profile of returns across the trading day, and its correlation heatmaps (Section 2).

- **Forecasting experiment.** Built 816 predictors and re-estimated Ridge, Lasso and Elastic Net once a year, each time on all earlier data. The out-of-sample R^2 is 0.171%, 0.151% and 0.151% (Section 4).

My extensions beyond the paper

Extension	Question it answers	Section
Paired trading-day block bootstrap	Are the differences between models real, or could they be luck?	4.2
Predictor ablation (removing groups of predictors and refitting)	Which groups of portfolios carry the predictive information?	5
Exact year-by-year breakdown of the accuracy gain	Is the advantage spread across years or concentrated in a few?	5
Trading evaluation with transaction costs	Do the forecasts translate into profits, and how much cost can they absorb?	6
Volatility-regime and time-of-day breakdown	Under which market conditions does the advantage appear?	6.5

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1 Key concepts

Returns and units

- **Simple return.** The percentage change in value over a period: $r_t = P_t/P_{t-1} - 1$, where P_t is the value at time t. A move from 100 to 101 is a return of 0.01 (1%).
- **Log return.** $\log(1 + r_t)$. Log returns add up over time, so the return over 15 minutes is the sum of the 15 one-minute log returns. For small returns, simple return and log return are nearly equal.
- **Basis point (bp).** One hundredth of a percent: 1 bp = 0.01% = 0.0001. The standard deviation of 15-minute market returns is roughly 18 bp.
- **Percentage point (pp).** The difference between two percentages. Going from 0.075% to 0.171% is an increase of 0.096 pp.
- **Annualized.** Rescaled to a one-year horizon so that numbers measured over different periods can be compared. A year has about 252 trading days. Average returns scale with time (multiply by 252). Volatility scales with the square root of time (multiply by $\sqrt{252}$).

Portfolios and factors

- **Portfolio.** A basket of stocks whose return is the weighted average of its members' returns.
- **The market.** A portfolio of essentially all US stocks, weighted by company size. Its next 15-minute return is what this project predicts.
- **Long and short.** Being *long* means owning an asset, so you gain when it rises. Being *short* means borrowing an asset and selling it, then buying it back later to return it, so you gain when it falls.
- **Factor portfolio.** A portfolio that is long one group of stocks and short another, chosen by a stock characteristic. Its return is the long group's return minus the short group's, so it measures how one kind of stock performs *relative to* another, with the overall market's movement largely cancelled.
- **FF6.** The five Fama–French factors plus momentum:
  - market;
  - size (small minus big companies);
  - value (cheap minus expensive);
  - profitability (robust minus weak);
  - investment (conservative minus aggressive asset growth);
  - momentum (recent winners minus recent losers).
- **Industry portfolios.** 48 portfolios that group stocks by business sector (for example banks, utilities, retail).
- **Characteristic portfolios and the "factor zoo".** Academic research has proposed hundreds of stock characteristics that seem to predict returns, and this crowd of factors is called the *factor zoo*. This project uses 218 of them. In the descriptive statistics, each is oriented by its published sign so that its long side is the side expected to earn more.

Risk and volatility

- **Volatility.** How much returns fluctuate, measured by their standard deviation. Higher volatility means bigger swings in either direction.
- **Realized variance (RV).** Variance measured from high-frequency data: the sum of squared returns over a period, $RV = \sum_t r_t^2$. Its square root is *realized volatility*.
- **Overnight return.** The price change between one day's close (16:00) and the next day's open (09:30). No regular trading happens in between, so it is treated separately from the 15-minute returns within the trading day.

Forecast evaluation

- **Out-of-sample.** A forecast is out-of-sample if the model was fitted without seeing the period being forecast. This is the honest way to measure predictive power.
- **Out-of-sample R^2.** The percentage reduction in squared forecast error compared with always predicting a return of zero:

$$R^2_{OOS} = 1 - \frac{\sum_t (y_t - \hat{y}_t)^2}{\sum_t y_t^2},$$

  where y_t is the realized return and $\hat{y}_t$ the forecast. A value of 0 means no better than predicting zero, and a negative value means worse. Stock prices are very close to unpredictable over

short horizons, so an R^2 of a few tenths of a percent is economically meaningful when it is earned on thousands of forecasts a year.

Trading terms

- **Notional.** The fixed amount of money a strategy trades with. Results scale in proportion to it, so they are reported per \$1 of notional.
- **Position.** w_t is how much of the market is held during interval t, as a multiple of the notional.
- **Profit and loss (P&L).** The gain or loss from a position: $\text{P\&L}_t = w_t \times r_t$, as a fraction of the notional.
- **Gross vs. net.** *Gross* means before trading costs and *net* means after.
- **Turnover.** The total amount traded, $\sum_t |w_t - w_{t-1}|$, in units of notional. Moving from long $(+1)$ to short (-1) trades 2 units: sell what you own, then short the same amount again.
- **Sharpe ratio.** Average return per unit of volatility; the standard measure of risk-adjusted performance:
$$\text{Sharpe} = \frac{\text{mean(daily P\&L)}}{\text{standard deviation(daily P\&L)}} \times \sqrt{252}.$$
Roughly, a Sharpe ratio of 1 means one year's average gain equals one standard deviation of the year's P&L.
- **Break-even cost.** The cost per unit traded at which total net P&L is exactly zero: total gross P&L divided by total turnover.
- **Hit rate.** The share of forecasts whose sign (up or down) was correct.
- **Maximum drawdown.** The largest fall in cumulative P&L from a previous peak: the worst loss an investor would have experienced from a high point.

2 Data and descriptive analysis

The descriptive sample covers **1996–2020**: 300 monthly files containing 2,454,420 one-minute timestamps on 6,295 trading days, each with a return for every portfolio. The 272 portfolios are 6 FF6 factors, 48 industry portfolios and 218 characteristic portfolios.

From one-minute to 15-minute returns. Each trading day starts with one observation at 09:30, which is the overnight return from the previous close. It is followed by 390 one-minute returns (09:31–16:00). The one-minute returns are combined within each day into 15-minute returns. For example, the interval labeled 09:45 contains the minutes 09:31–09:45. The overnight observation stays separate. Shortened sessions, such as the day after Thanksgiving, are kept as observed, and missing intervals are not filled with zeros. This gives **169,518 15-minute observations**, including the overnight ones. The descriptive statistics use 15-minute log returns, and the forecasts later use simple returns.

Portfolio statistics

Averages of portfolio-level statistics within each group, in percent, excluding overnight returns:

Average statistic (%)	Market	FF6	Industry 48	Characteristic 218
Annual log return	1.07	3.74	1.32	2.93
Annualized realized volatility	14.42	8.59	19.62	8.87
Relative jump measure	2.76	4.75	5.21	4.69
First-order autocorrelation	3.56	6.11	3.28	4.00
Correlation with market	100.00	0.68	67.32	−9.28

How each row is computed, for a portfolio's 15-minute log returns r_t over 25 years:

- **Annual return:** $\sum_t r_t/25$.
- **Annualized volatility:** $\sqrt{RV/25}$, where $RV = \sum_t r_t^2$.
- **Relative jump measure:** the share of variance coming from sudden large price moves ("jumps") rather than continuous movement, estimated as $(RV - BP)/RV$. Here $BP = \frac{\pi}{2}\sum_t |r_t\, r_{t-1}|$ is *bipower variation*, a variance estimate that is largely unaffected by jumps.
- **First-order autocorrelation:** $\sum_t r_t r_{t-1}/RV$, which measures how much one interval's return resembles the next. Positive values mean some continuation, which is the basic reason prediction might be possible.
- **Correlation with the market:** the same cross-product normalized by both portfolios' realized volatilities.

These are realized-moment statistics rather than centered sample correlations or formal jump tests.

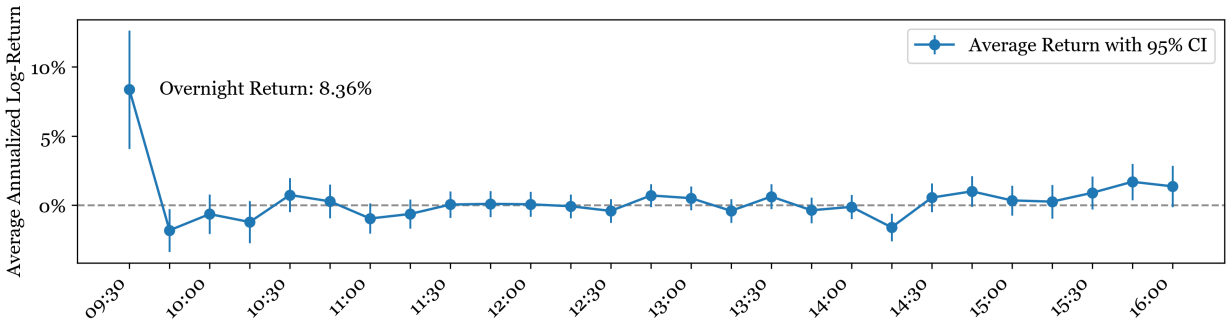


Figure 2: Intraday market log-return contributions. Average annual market return earned in each 15-minute interval, 1996–2020, with 95% error bars of half-width $1.96\sqrt{\sum r^2}/25$.

The overnight return contributes 8.36% per year, against about **1.07%** from all regular-session intervals combined. Most of the stock market's long-run gain arrives while the market is closed.

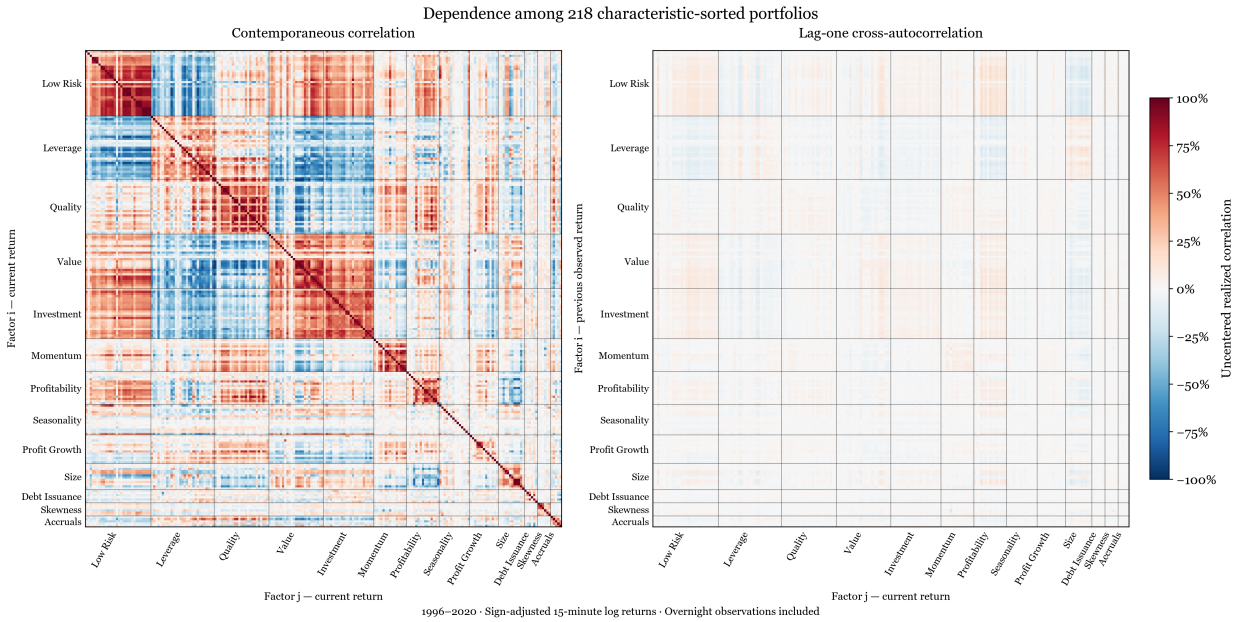


Figure 3: Dependence among characteristic portfolios.

The left panel shows how strongly the 218 characteristic portfolios move together *in the same* 15-minute interval. The right panel shows how strongly each portfolio's return predicts each other portfolio's return in the *next* interval (rows are lagged factors, columns are current factors). Both panels include overnight returns. Movement in the same interval is strong, and next-interval dependence is much weaker. That pattern motivates **regularization**: many weak, correlated signals have to be combined to avoid overfitting.

3 Forecasting design

Target and predictors

The **target** y_t is the market's simple return over one 15-minute interval of the regular trading session (09:30–16:00, US Eastern time). The forecast for that interval is made at its start, using only returns observed up to then.

The **predictors** describe how each of the 272 portfolios has behaved recently, at three horizons:

Predictor	Definition
`lag15`	The portfolio's previous 15-minute return
`avg60`	The average of the portfolio's previous four 15-minute returns (one hour)
`prevday`	The average of the portfolio's 15-minute returns on the previous trading day

That gives $272 \times 3 =$ **816 predictors**. Overnight returns are removed before any lag or average is computed, so the hour and day averages reach back across days without using the overnight gap. The resulting dataset has 137,031 intervals starting 2000-01-04, of which **111,084 intervals on 4,280 trading days** (2004–2020) are forecast out-of-sample.

Forecasting models

All three candidate models are linear regression models, but they penalize large coefficients differently. This *regularization* trades a little bias for much less overfitting when there are many noisy, correlated predictors.

- **Ridge.** Shrinks all coefficients toward zero smoothly: $\min_\beta \frac{1}{2n}\|y - X\beta\|^2 + \frac{\lambda}{2}\|\beta\|_2^2$.
- **Lasso.** Sets many coefficients exactly to zero, which selects predictors: $\min_\beta \frac{1}{2n}\|y - X\beta\|^2 + \alpha\|\beta\|_1$.
- **Elastic Net.** Mixes the two penalties: $\alpha\rho\|\beta\|_1 + \frac{\alpha(1-\rho)}{2}\|\beta\|_2^2$.

Model	Penalty values searched
Ridge	100 values of λ from e^{-1} to e^{8}, evenly spaced on a log scale
Lasso	100 values of α from e^{-6} to e^{-1}
Elastic Net	50 values of α from e^{-6} to e^{1}, each with 50 values of ρ from 0.05 to 0.95

Annual re-estimation without look-ahead

Each test year is forecast by models fitted only on earlier data:

1. **Split history.** All data from 2000 up to the end of the previous year are split by calendar time. The first 80% is *training* data and the last 20% is *validation* data. For example, for 2004, training runs to 2003-03-13 and validation from 2003-03-14 to 2003-12-31: 20,712 training, 5,235 validation and 6,541 test intervals.

2. **Standardize.** Each predictor and the target are rescaled to mean 0 and standard deviation 1, using the training period's means and standard deviations only. For the target, that is $z = (y - \bar{y}_{\text{train}})/s_{\text{train}}$.

3. **Tune.** Each candidate penalty is fitted on training data and scored by its mean squared error on the validation data, in return units. The penalty with the lowest validation error is kept.

4. **Forecast.** The selected model, fitted on training data only, forecasts every interval of the test year. It predicts the *standardized* target $\hat{z}$, so each forecast is converted back to a return:

$$\hat{y} = \bar{y}_{\text{train}} + s_{\text{train}}\,\hat{z}.$$

   Each forecast is therefore the training-period average return plus an adjustment driven by the predictors. When every predictor equals its training average, the adjustment is zero and the forecast is just the training average.

This repeats for every year from 2004 to 2020, so all 111,084 forecasts are out-of-sample.

Pooled R^2

Accuracy is summarized by **pooled** R^2, computed once from all 111,084 test intervals together rather than averaged across years. Pooled R^2 equals a weighted average of the yearly R^2 values, with each year weighted by its sum of squared returns. Volatile years therefore carry more weight.

4 Model comparison

4.1 Forecast accuracy

All three models forecast exactly the same 111,084 test intervals.

Model	Out-of-sample R^2 (%)
Ridge	0.1714
Lasso	0.1511
Elastic Net	0.1505
Always predict zero (benchmark)	0
Always predict the training-period mean	−0.0011

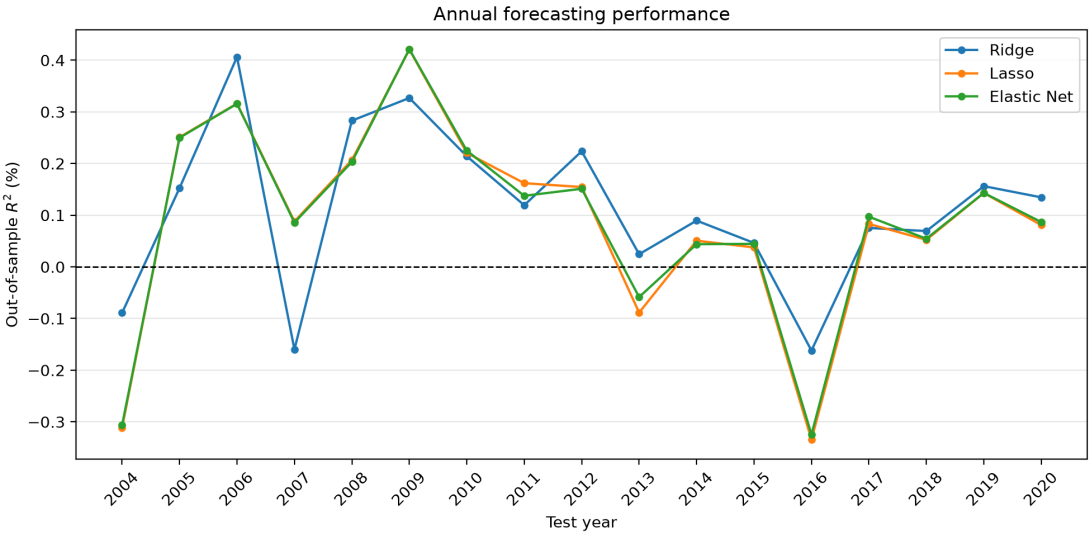


Figure 4: Annual model accuracy.

- **All three models beat the zero forecast** over the full test period, and the values match the paper to within 0.003 percentage points.
- **Ridge has the highest point estimate.** Whether its lead is real or luck is the question for Section 4.2.
- **Lasso and Elastic Net are almost the same model here.** Their forecasts have correlation 0.998, and the root-mean-square difference between them is 0.037 bp.
- **The historical average does not help.** Predicting the training-period mean does slightly *worse* than predicting zero, because at a 15-minute horizon the average return is tiny and too noisy to estimate.
- **Accuracy varies a lot from year to year.** Ridge's annual R^2 ranges from -0.16% (2016) to 0.41% (2006) and is negative in 3 of 17 years. Pooled R^2 is driven most by volatile years such as 2008.

4.2 Uncertainty analysis

The values above come from one particular 17-year history. A different history would give somewhat different numbers. The **bootstrap** estimates how much they could plausibly vary by resampling the observed trading days many times and recomputing each statistic. It works in four steps.

Step 1: summarize each trading day with three kinds of sums. For trading day d, with realized returns y_t and model m's forecasts $\hat{y}_{m,t}$ in its intervals t:

Daily sum	Formula	Meaning
Interval count	n_d	Number of 15-minute intervals that day (usually 26)
Squared returns	$S_d = \sum_{t \in d} y_t^2$	The squared error a zero forecast would make that day
Squared errors of model m	$SSE_{m,d} = \sum_{t \in d} (y_t - \hat{y}_{m,t})^2$	Model m's squared forecast error that day

Pooled R^2 can be computed exactly from these daily sums:

$$R^2_m = 1 - \frac{\sum_d SSE_{m,d}}{\sum_d S_d}.$$

The 111,084 intervals therefore reduce to 4,280 daily rows. Each day stays together as one unit, and a day with more intervals automatically carries more weight.

Step 2: draw a new history from blocks of days. Pick a random starting day, take it and the next 4 trading days (a 5-day block), and repeat until 4,280 days have been drawn. A block that runs past the last day wraps around to the first. A day can be drawn more than once and then counts each time.

Step 3: recompute. Use the formula above on the drawn days to get each model's R^2 and the differences between models.

Step 4: repeat. Repeat Steps 2–3 5,000 times. The 2.5th and 97.5th percentiles of the 5,000 values form the **95% bootstrap percentile interval**.

Why blocks of days, not single intervals? The bootstrap assumes the resampled pieces are independent. Single intervals are not: a model that is wrong at 10:00 on a volatile day is often wrong at 10:15 too, and volatile days cluster over weeks. Resampling single intervals would break those links and make the intervals too narrow. Whole days keep within-day dependence, and multi-day blocks keep short-run volatility clustering. Results are reported for 5-day, 10-day and 20-day blocks; similar intervals across lengths mean the choice of length does not matter much.

Why "paired"? In every draw, all models are scored on the *same* drawn days, and their difference is computed inside that draw. Most of each model's squared error is the unpredictable

part of the return, which all models share. When one model has an unlucky draw, the others usually do too. Pairing cancels this shared noise:

$$\mathrm{Var}(R^2_A - R^2_B) = \mathrm{Var}(R^2_A) + \mathrm{Var}(R^2_B) - 2\,\mathrm{Cov}(R^2_A, R^2_B).$$

The covariance is large, so a difference is far less uncertain than either R^2 on its own.

What the intervals do not cover. The models are not refitted in each draw, so the intervals measure uncertainty from *which days occurred*, given the saved forecasts.

Model	R^2 **(%)**	**95% interval (%)**
Ridge	0.1714	[0.102, 0.239]
Lasso	0.1511	[0.083, 0.218]
Elastic Net	0.1505	[0.085, 0.216]

Difference in R^2	**Estimate (pp)**	**95% interval (pp)**
Ridge − Lasso	0.020	[−0.011, 0.050]
Ridge − Elastic Net	0.021	[−0.010, 0.050]
Lasso − Elastic Net	0.0006	[−0.003, 0.004]

Intervals use 5-day blocks; 10- and 20-day blocks give the same conclusions.

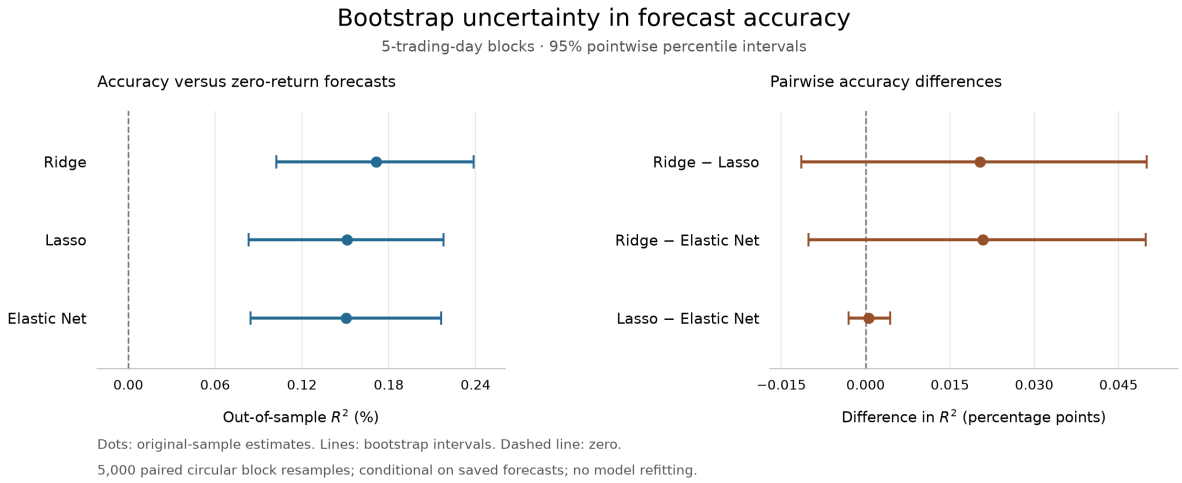


Figure 5: Model accuracy and paired uncertainty.

Interpretation.

- **The models genuinely beat the zero forecast.** Every model's interval lies entirely above zero at all three block lengths.
- **No model is clearly the best.** Every pairwise interval includes zero, so Ridge's higher point estimate could be luck. The accurate statement is that Ridge does *at least as well* as Lasso and Elastic Net.
- **A plausible reason Ridge is not worse:** the predictive information appears to be spread thinly over many correlated predictors. Ridge keeps every predictor with a small weight, which averages out noise, whereas Lasso tries to pick a few and drops the rest.

Ridge is used for the rest of the analysis. With all 816 predictors it is called **Full**.

5 Which predictors matter?

Section 4 shows that the full model predicts better than zero. This section asks *where that accuracy comes from*, using three analyses of Ridge:

Analysis	What is done	Question it answers
Predictor ablation	Refit Ridge each year on a reduced predictor set, retuning its penalty: **market only** (the market's own 3 predictors) and **without characteristics** (the 54 FF6 and industry portfolios, 162 predictors).	Do the other 271 portfolios improve on the market's own past returns? Do the 218 characteristic portfolios add anything beyond industry and FF6?
Regularization sensitivity	Refit the market-only model with a wider range of penalties, including weaker ones and none at all.	Did the market-only model lose because its penalty choices were too limited, rather than because it lacks information?
Annual contributions	Split Full's total accuracy advantage into one piece per year, where the pieces add up exactly to the total.	Is Full's advantage spread across many years, or does it come from a few?

All intervals use the same paired block bootstrap method as Section 4.2.

Removing predictor groups

Each reduced set gets fresh annual Ridge fits with its own tuned penalty, on exactly the same rows and splits:

Ridge predictors	Count	R^2 (%)	95% interval (%)
Full: all 272 portfolios	816	0.1714	[0.102, 0.239]
Without characteristics: industry + FF6	162	0.1665	[0.073, 0.254]
Market only (original penalty grid)	3	0.0751	[−0.004, 0.147]
Market only (expanded penalty grid)	3	0.0634	[−0.022, 0.139]

Comparison	Difference (pp)	95% interval (pp)
Full − market only (original grid)	0.096	[0.038, 0.159]
Full − market only (expanded grid)	0.108	[0.046, 0.176]
Full − without characteristics	0.005	[−0.042, 0.051]

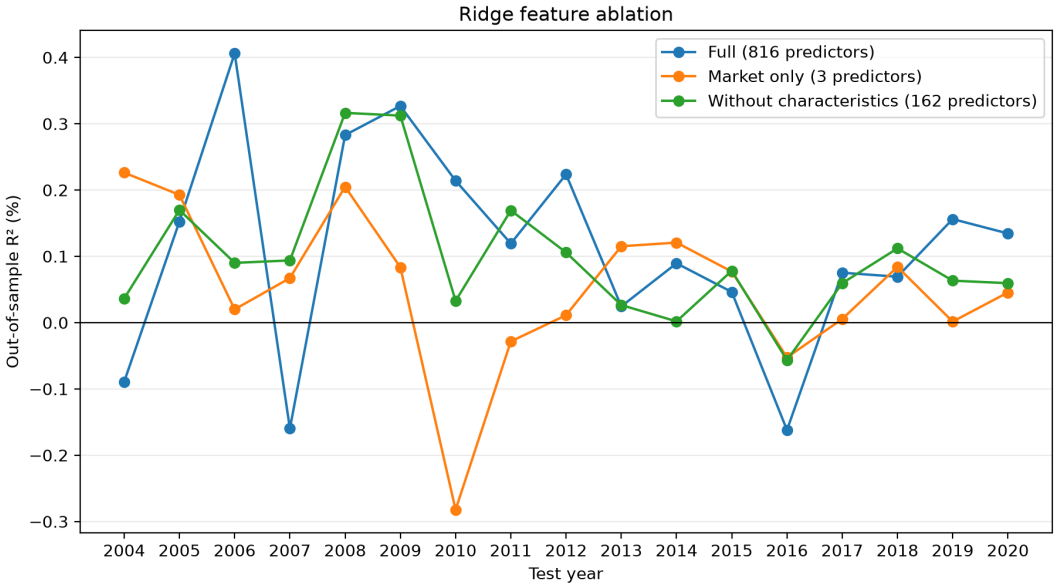


Figure 6: Feature-set accuracy under the original penalty grid.

Interpretation.

- The other portfolios clearly add information beyond the market's own past returns. This holds at all three block lengths; for example, against the expanded-grid market-only model the 10-day interval is [0.048, 0.174] and the 20-day interval is [0.046, 0.176].
- Adding the 218 characteristic portfolios on top of industry and FF6 portfolios gives no detectable further gain. That does not prove they are useless; the data simply cannot separate the two.

Was the market-only benchmark handicapped?

In 7 of 17 years (2004, 2005, 2009, 2010, 2011, 2015, 2016), the market-only model chose the *smallest* penalty on its grid ($\lambda \approx 0.368$). It might have preferred an even weaker penalty if one had been available. To test this, the grid was expanded:

- The 100 original values were kept.
- 60 smaller values were added (`geomspace(1e-6, exp(-1), 60, endpoint=False)`).
- Zero penalty was added (ordinary least squares).

All 161 candidates were scored on validation data only. The seven affected years moved to weaker penalties (zero in 2009), and the other ten years were unchanged.

The expanded benchmark did *worse* out of sample, by −0.012 pp with interval [−0.022, −0.003], even though it fit the validation data at least as well. Both versions are reported, and test results were not used to choose between them. **Full's advantage does not come from a handicapped benchmark.**

Which years produce the advantage?

Each year's contribution is the error reduction in that year divided by the *whole period's* sum of squared returns:

$$\text{contribution}_{\text{year}} = 100 \times \frac{SSE^{\text{market only}}_{\text{year}} - SSE^{\text{full}}_{\text{year}}}{\sum_{\text{all years}} y_t^2}.$$

Here SSE is the sum of squared forecast errors.

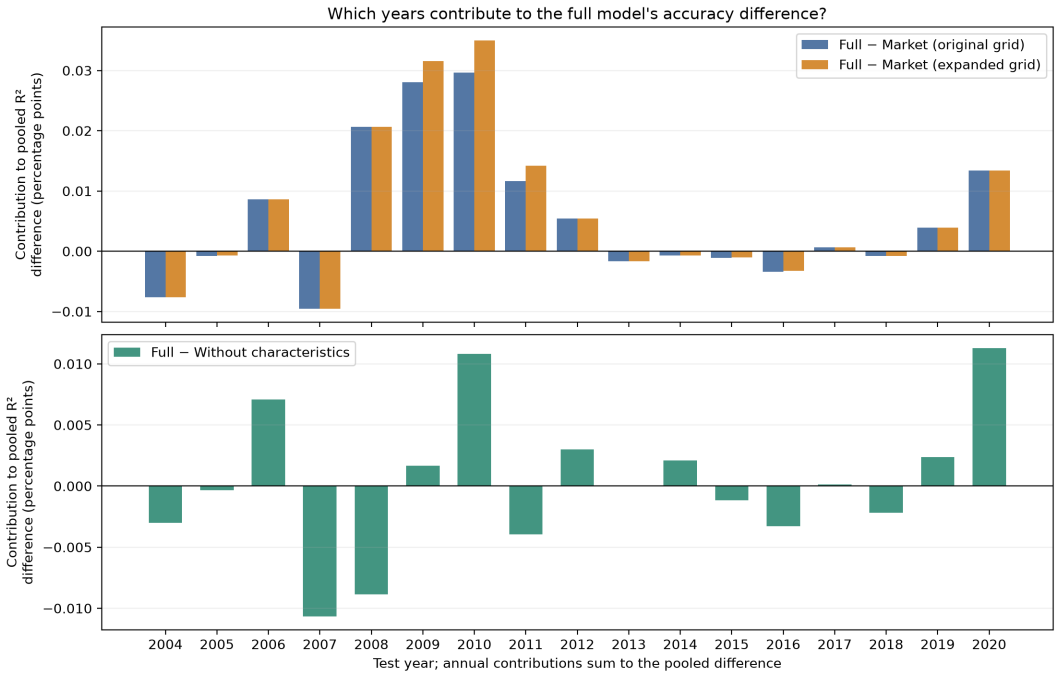


Figure 7: Annual contributions to pooled accuracy differences.

- Full beats market-only in 9 of 17 years.
- Against the expanded-grid benchmark, **2008–2010 contribute 0.087 pp, about 81% of the 0.108 pp total**. All other years together contribute a net 0.021 pp.
- Against the without-characteristics model, Full wins in 8 of 17 years, and gains and losses roughly cancel.

These years were identified after looking at the results, so this is descriptive and not a causal claim. Section 6.5 tests the obvious explanation: the 2008–2010 crisis was very volatile.

6 From forecasts to trading

An R^2 of 0.17% is hard to interpret in money terms. This section turns the saved forecasts into simple trading rules and measures their profits, costs and risk. Nothing is refitted.

Setup (code: `strategy_evaluation.py`):

- **Models.** Full, compared with Market only (expanded penalty grid).
- **Timing.** Each position is opened at the forecast time and held for the next 15 minutes. All positions are closed at the end of every day (nothing is held overnight). Trades are assumed to fill instantly at the forecast time.
- **Traded asset.** The market portfolio itself. It cannot be bought directly, so the results are a gross proxy for an index fund such as SPY or an index future.
- **Uncertainty.** Every statistic comes with the same 5-day block bootstrap as above, with 5,000 resamples shared by all statistics. Intervals for 10- and 20-day blocks are in the saved tables and are similar.

6.1 Do larger forecasts come before larger returns?

The forecast-decile chart (Figure 1) sorts each year's forecasts into ten equal groups and averages the realized return in each group. Sorting within years stops the extreme deciles from being filled with volatile crisis periods.

- Full: realized returns rise from −1.35 **bp** (lowest decile) to +1.35 **bp** (highest). The top-minus-bottom spread is **2.70 bp** [2.08, 3.31].
- Market only: the spread is 2.16 bp [1.55, 2.78].
- In both, realized returns track the forecasts closely, so the forecasts are roughly calibrated, not just correctly ordered.

Hit rate (correct direction)	**Estimate**	**95% interval**
Always predict "up"	51.8%	[51.5, 52.1]
Full	50.1%	[49.7, 50.4]
Market only	49.2%	[48.9, 49.5]

Hit rate is not the right measure here. The market rose in 51.8% of intervals with a nonzero return, so always guessing "up" scores better than either model. The models make money anyway (next section) because they tend to be right on *large* moves and wrong on small ones, and profit depends on the size of moves, not only their direction.

6.2 A simple trading rule

Rule (sign). Go long \$1 when the forecast is positive and short \$1 when it is negative: $w_t = \text{sign}(\hat{y}_t)$.

Daily net P&L at a cost of c per unit traded is

$$\text{net P\&L}_d = \sum_{t \in d} w_t \, r_t \; - \; c \times \text{turnover}_d,$$
$$\text{turnover}_d = |w_{\text{first}}| + \sum_{t \in d} |w_t - w_{t-1}| + |w_{\text{last}}|.$$

The turnover includes opening the first position and closing the last one each day.

Strategy	**Forecasts used**	**Trading rule**
Long only (session)	None	Buy \$1 of the market at the 09:30 open, hold it through the day, and sell at the 16:00 close; a benchmark for simply holding the market during trading hours.
Full: sign	Full	Long \$1 when the forecast is positive, short \$1 when it is negative.
Market only: sign	Market only (expanded grid)	The sign rule applied to Market only's forecasts.

Strategy (gross, 2004–2020)	**Annual return**	**Annual volatility**	**Sharpe**	**Max drawdown**	**Average daily turnover**
Long only during the session	2.5%	15.2%	0.17 [−0.27, 0.62]	54.4%	2.0
Market only: sign	27.8%	14.3%	1.94 [1.50, 2.37]	15.6%	18.3
Full: sign	**31.3%**	**14.6%**	**2.15 [1.71, 2.58]**	**13.9%**	**16.8**

Returns are on \$1 of notional, summed rather than compounded, before costs. "Average daily turnover" is in units of notional.

The gross Sharpe ratio of 2.15 looks impressive, but the rule trades about 17 times its capital every day. Its **break-even cost is 0.74 bp** per unit traded [0.57, 0.91]:

Cost per unit traded	**0 bp**	**0.25 bp**	**0.5 bp**	**1 bp**	**2 bp**
Full: sign, net Sharpe	2.15	1.42	0.69	−0.74	−3.48
Market only: sign, net Sharpe	1.94	1.13	0.33	−1.23	−4.20

For comparison, SPY's quoted buying and selling prices usually differ by one cent. Each trade pays about half of that gap. At SPY prices of roughly \$70 to \$370 over 2004–2020, half a cent is about 0.14–0.7 bp per unit traded, before fees and market impact. So the edge is thin even for a low-cost trader, and it disappears at 1 bp.

6.3 Trading only on strong forecasts

Small forecasts are mostly noise, yet trading them costs as much as trading large ones. The **threshold rules** are based on the sign rule but take a position only when the absolute forecast exceeds a percentile of the absolute forecasts over the **previous 20 trading days** (fewer at the very start of 2004). Using only past forecasts keeps the rule free of look-ahead. The **linear** rule instead sizes positions in proportion to the forecast, scaled by the average forecast size over the previous 20 trading days.

Strategy using Full forecasts	**Avg. abs. position**	**Avg. daily turnover**	**Gross Sharpe**	**Break-even cost**	**Net Sharpe at 0.5 bp**	**Net Sharpe at 1 bp**
Sign	1.00	16.8	2.15	0.74 bp	0.69	−0.74
Above trailing median	0.50	12.1	2.35	0.86 bp	0.98	−0.37
Above trailing 75th percentile	0.25	7.3	2.08	0.94 bp	0.98	−0.12
Above trailing 90th percentile	0.10	3.4	1.66	**1.10 bp**	0.90	0.15
Linear	1.01	25.1	2.22	0.85 bp	0.91	−0.40

Average absolute position is the average of $|w_t|$ over all intervals. For the threshold rules it equals the proportion of intervals with a position. For the linear rule the position size varies, but averages about 1.

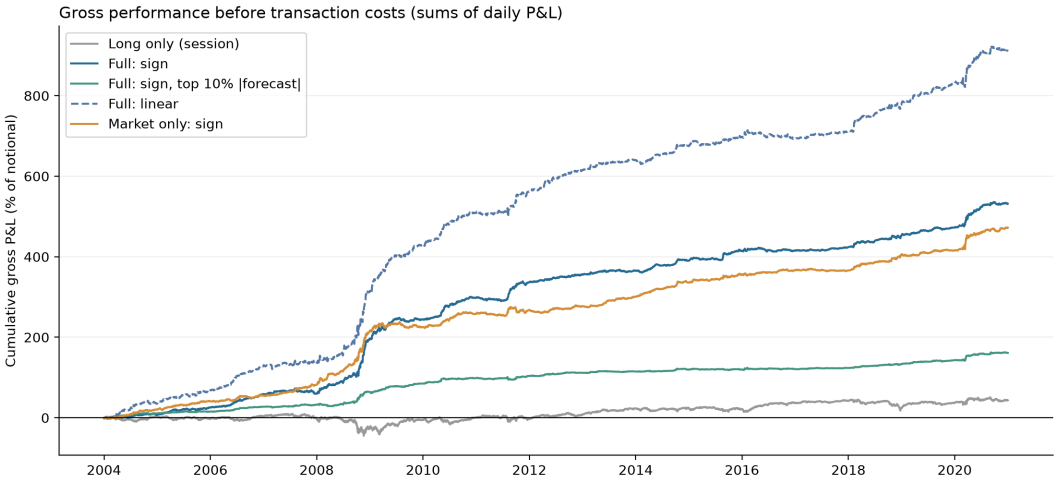


Figure 8: Cumulative gross P&L. Each line is the running total of daily gross P&L from 2004-01-02, in percent of \$1 of notional, summed rather than compounded.

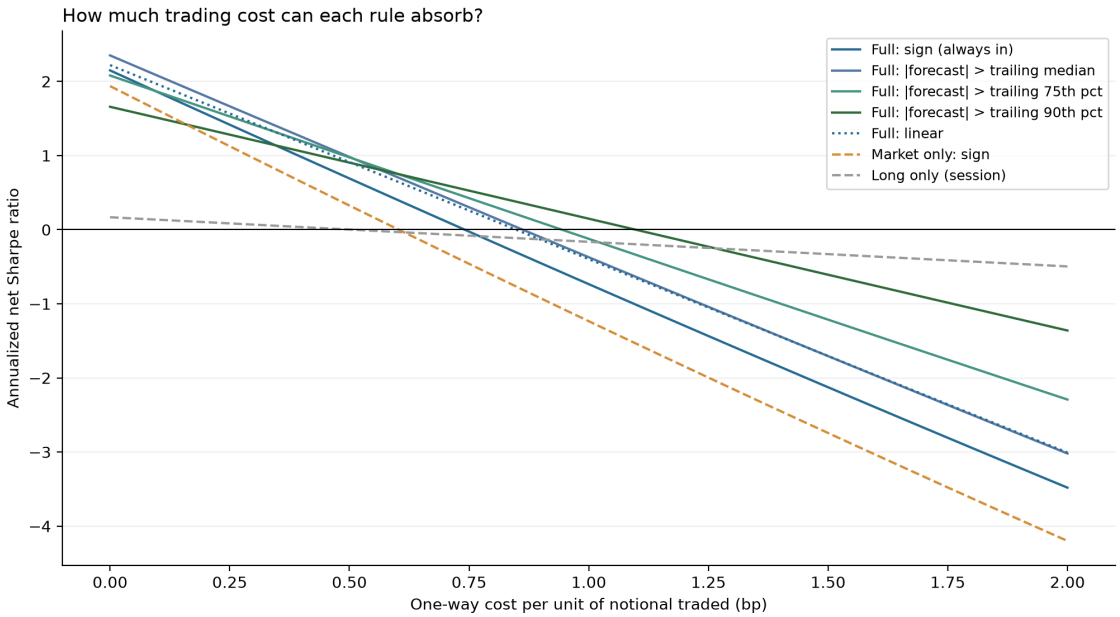


Figure 9: Net Sharpe ratio against trading cost.

Trading only above the median raises the gross Sharpe ratio slightly (2.35 vs 2.15), but stricter thresholds lower it (2.08 and 1.66). In exchange, fewer trades make the strategy cheaper to run. The 90th-percentile rule trades about a fifth as much as the sign rule and keeps the highest net Sharpe once costs pass about 0.65 bp.

6.4 How close is trading performance to what R^2 implies?

Campbell and Thompson (2008) link forecast accuracy to the best attainable Sharpe ratio. Assume the forecast is the true expected return and volatility is constant. Then a position proportional to the forecast has a per-interval Sharpe ratio of $\sqrt{R^2/(1-R^2)}$. Multiplying by $\sqrt{\text{intervals per year}}$, about $\sqrt{6{,}540}$, annualizes it. Under the same assumptions, no other way of trading on the forecast can do better, so this is also an upper bound for the sign rule, which is shown for reference:

Model	R^2 (%)	Implied Sharpe ceiling	Realized: linear	Realized: sign
Full	0.171	3.35 [2.59, 3.96]	2.22 [1.77, 2.66]	2.15 [1.71, 2.58]
Market only	0.063	2.04 [0, 3.02]	1.11 [0.58, 1.74]	1.94 [1.50, 2.37]

Full's linear rule captures about **two-thirds of the theoretical ceiling** (2.22/3.35). The gap is expected: real forecasts are noisy and volatility changes over time. These ceilings are before costs, which is exactly where the practical difficulty lies.

6.5 When does the full model help?

Section 5 found that Full's advantage over Market only comes mostly from 2008–2010 (about 81% of the total). Why those years? There are two possible explanations:

- **Crisis-specific.** Something special happened during the 2008–2010 financial crisis that made the 272 portfolios useful for forecasting, and it happened only in that crisis. If this is true, the advantage was a one-time event and should be gone after 2010.
- **Volatility-driven.** The 272 portfolios help whenever the market is volatile, in any year. 2008–2010 had more volatile days than any other period, so it shows the largest advantage. If this is true, the advantage should also appear on volatile days in other years, such as 2011 or 2020.

The difference matters for using the model today:

	Crisis-specific	**Volatility-driven**
Is the advantage still useful after 2010?	No	Yes, on volatile days

To tell them apart, each test day is classified by **ex-ante volatility**, a volatility estimate built only from data available before the day opens:

$$\sigma_d = \sqrt{252 \times \tfrac{1}{5}\textstyle\sum_{k=1}^{5} RV_{d-k}}, \qquad RV_d = \sum_{t \in d} r_t^2.$$

Days are split into equal thirds (terciles) by σ_d, with cut-offs at 7.7% and 11.3% annualized. These figures cover only the trading session, so they are lower than the usual quoted market volatility, which includes overnight moves. They are also below the 14.42% in the portfolio statistics table (Section 2), because they are boundaries between groups of days, not an average: most days are calm, and a few very volatile days (for example in 2008) pull the average up. The cut-offs use the whole sample, so this split describes the data rather than forming a trading filter.

Volatility tercile	**Share of realized variance**	**R^2 gain within tercile (pp)**	**Contribution to the 0.108 pp total (pp)**
Low	8%	−0.048 [−0.167, 0.068]	−0.004 [−0.014, 0.006]
Medium	15%	−0.010 [−0.112, 0.094]	−0.001 [−0.017, 0.014]
High	77%	**0.148 [0.073, 0.234]**	**0.113 [0.056, 0.178]**

Contribution of high-volatility days (pp)	**2004–2007**	**2008–2010**	**2011–2020**
Estimate	−0.001	0.083	**0.032**
95% interval	[−0.013, 0.009]	[0.033, 0.142]	**[0.004, 0.062]**

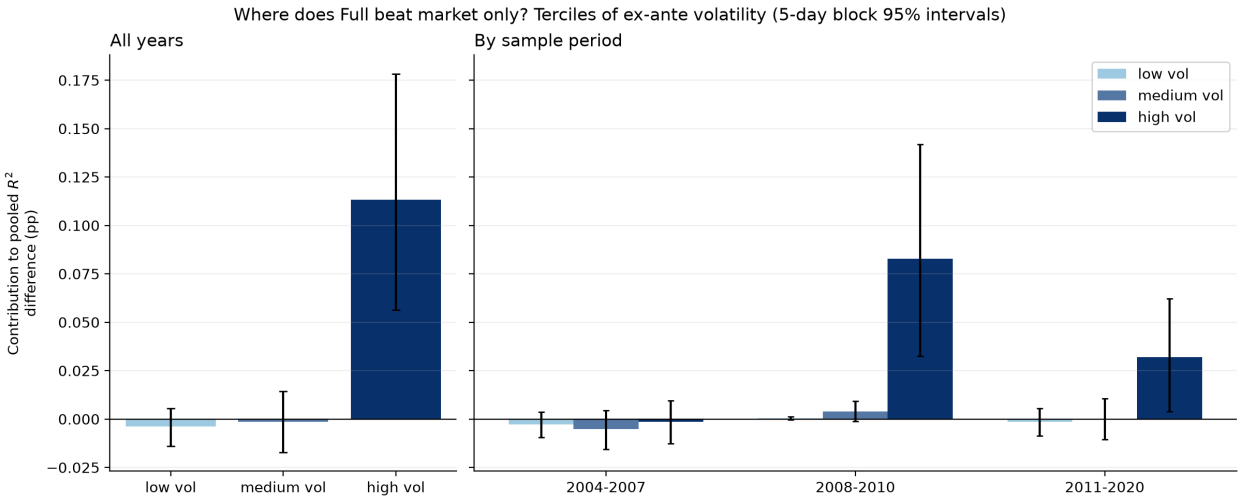


Figure 10: Contributions by volatility regime.

Interpretation.

- **All of Full's advantage comes from high-volatility days.** On calm days it is no better than the market-only model.
- **The effect is not only a crisis effect.** High-volatility days after 2010 still show a positive contribution whose interval excludes zero, and a within-tercile R^2 gain of 0.103 pp [0.013, 0.199], though it is smaller than in 2008–2010.

- High-volatility days are a third of the days but carry 77% of the realized variance, which is their weight in pooled R^2.
- The volatile days of 2004–2007 show no gain, so volatility is not the whole story.

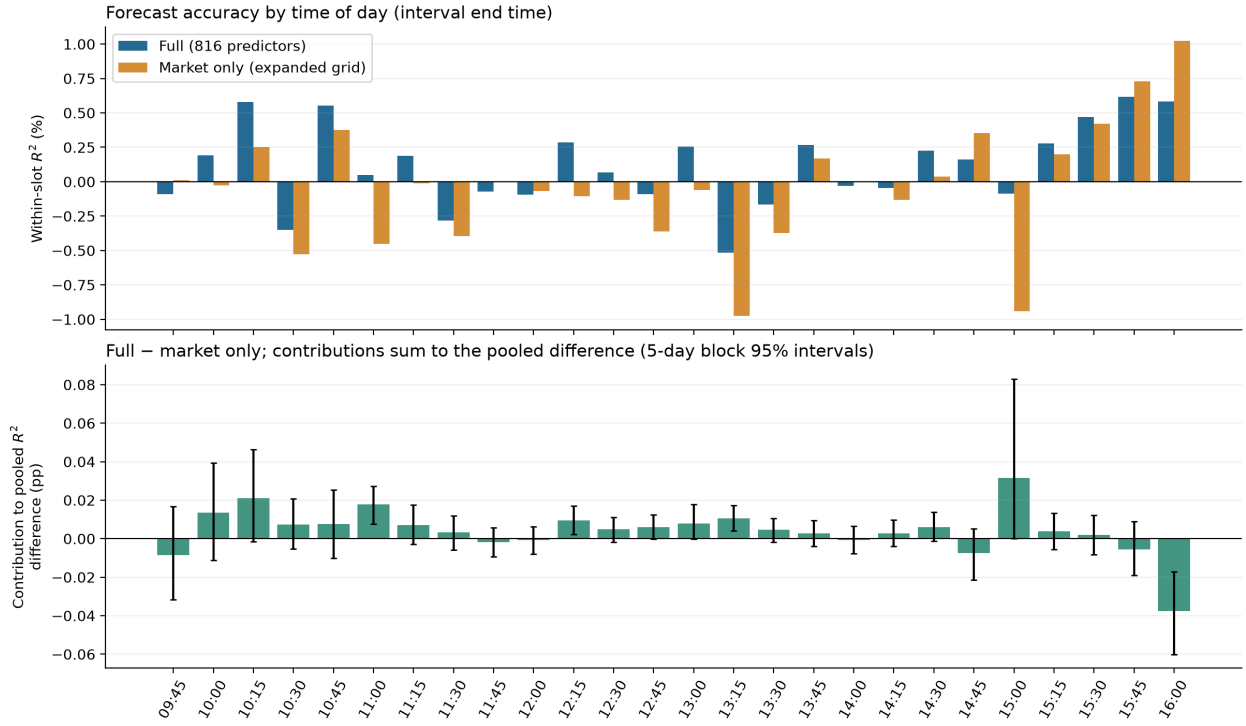


Figure 11: Contributions by time of day.

Time of day. Full beats market-only in 19 of the 26 intervals. The first hour contributes 0.034 pp of the 0.108 pp total, the middle of the day 0.112 pp, and the last hour −0.037 pp. Almost all of that loss comes from the **last interval (15:45–16:00)**, where Full is significantly *worse* than market-only: −0.038 pp [−0.060, −0.017]. One possible reason is that the final minutes are dominated by end-of-day order flow, which past portfolio returns would not capture; this explanation is not tested here. With 26 intervals tested, a few intervals will look significant by chance, so single-interval results should be read cautiously.

7 Limitations

Scope and reproduction differences

- **Partial reproduction.** Observed factor returns are used to predict observed market returns. The paper's separate modelling of continuous and jump components, its full model ensemble and its SPY trading strategy are not reproduced. Section 6 is a simpler, separately designed evaluation.

Statistical caveats

- **The 95% intervals are approximate.** They come from the block bootstrap in Section 4.2, and there are three things they do not include:
  - The models are not refitted in each resample. They are fitted once for each test year, and their forecasts are saved. Each resample draws a new set of test days and recomputes R^2 from the same saved forecasts. So the intervals answer "with these fitted models, how much would R^2 change if different days had occurred?" They do not answer "if the

training data had been a little different, how much would the coefficients, and so the forecasts, have changed?" That second source of uncertainty is not included, so the true uncertainty is somewhat larger than the intervals show.

  - Each interval is a 95% interval on its own. When many intervals are shown (for example, the 26 times of day), a few will exclude zero by chance.

- **Some analyses were designed after seeing results.** Every forecast uses only past data. But the later analyses (removing predictor groups, the wider penalty search, the trading rules and the volatility groups) were designed after seeing the first results for 2004–2020. So 2004–2020 is not completely new data for them. Testing on data after 2020 would be the cleanest check.
- **The volatility groups use future information.** The boundaries between low-, medium- and high-volatility days are computed from all days in 2004–2020, so this split describes the past and cannot be used for trading in real time. The 2008–2010 period was also chosen after seeing the yearly results.

Trading caveats

- **Not directly tradable.** The market portfolio is a proxy. An index fund or future would track it closely but not exactly.
- **Simplified execution.** The evaluation assumes zero latency: trades fill at the forecast time at the 15-minute return's starting price. Costs are a flat amount per unit traded, without market impact, fees, borrowing costs or capacity limits.

References

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- Campbell, J. Y., and Thompson, S. B. (2008). Predicting Excess Stock Returns Out of Sample: Can Anything Beat the Historical Average? Review of Financial Studies, 21(4), 1509–1531.